

## Graphic detail

Jul 17th 2021 edition

Fitbit for purpose

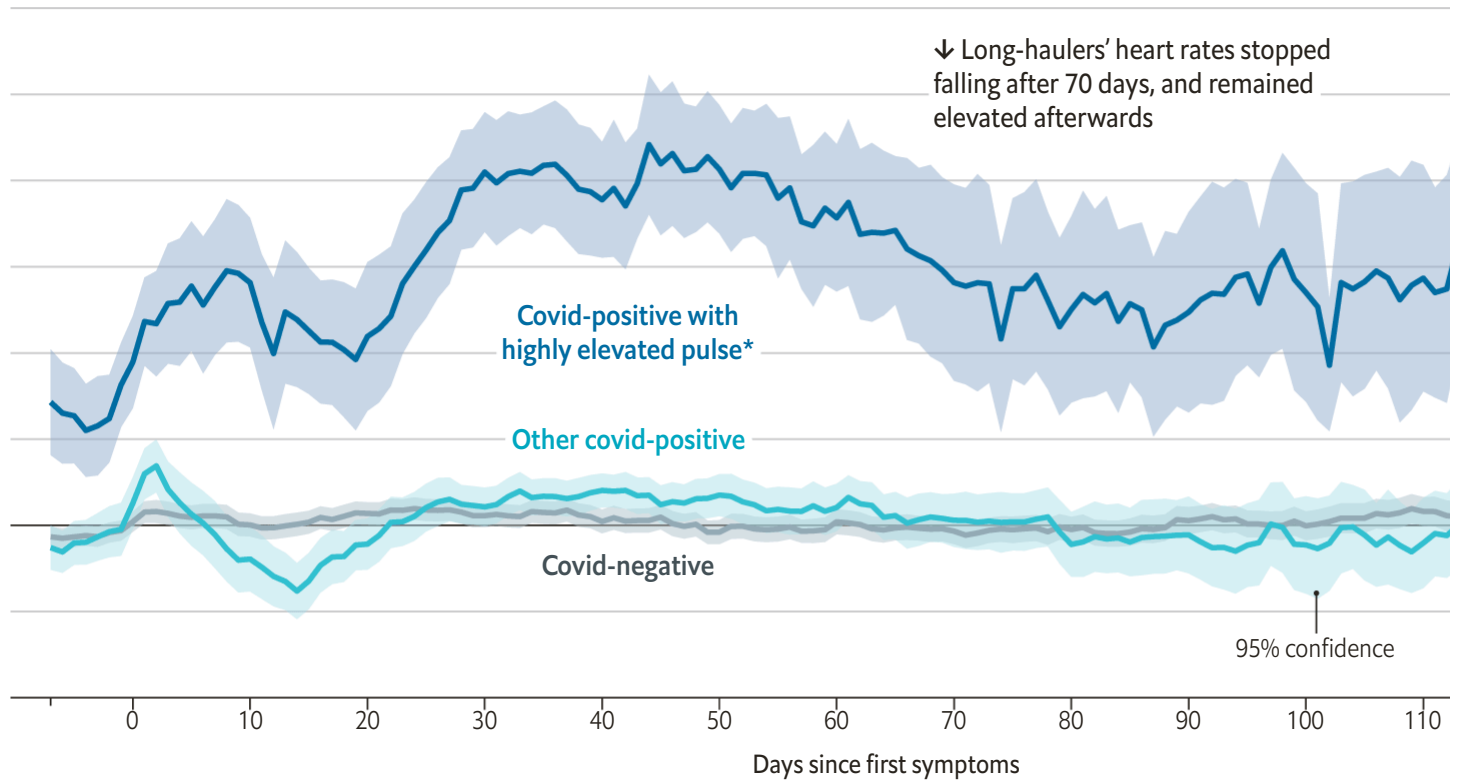
# A new study using wearable devices could help to define long covid

14% of covid-19 patients have elevated heart rates for months after infection

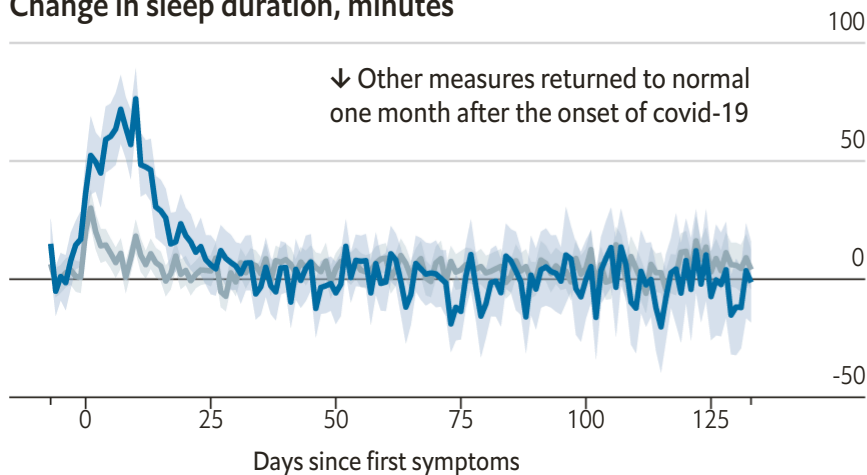
## 875 people who reported a respiratory illness and were tested for covid-19

United States, March 25th 2020-January 24th 2021

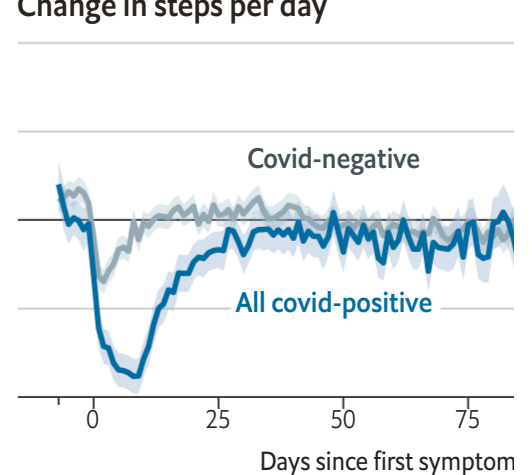
### Change in resting heart rate, beats per minute



### Change in sleep duration, minutes



### Change in steps per day



\*Average resting heart rate 4-8 weeks after onset of symptoms at least 5

THE GRAVEST costs of the covid-19 pandemic, in lives lost and economies paralysed, have been plain to see. But one of its most durable consequences is also among the hardest to measure: the plight of “long-haulers” with chronic symptoms.

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Doctors have not yet agreed on a definition of long covid. Some use it to describe symptoms lasting at least four weeks; others for six months or more. Nor is there consensus on which symptoms qualify. Most accounts cite exercise intolerance or shortness of breath; changes to involuntary functions like heart rates; or cognitive impairment (“brain fog”). But the severity needed for a diagnosis is open to debate.

Share of participants with respiratory symptoms, %



\*Average resting heart rate 4-8 weeks after onset of symptoms at least 5 beats per minute above pre-covid level

Covid-19 is too new for long-run studies of its impact to be complete. However, a new paper by researchers at Scripps Research Translational Institute and CareEvolution, a health-care firm, suggests that heart rates can distinguish some long-haulers from patients who recover fully.

In March 2020 Scripps began obtaining data from users of wearable fitness trackers. So far, 38,000 people have enrolled. The study covers 875 participants who had symptoms of respiratory illness and were tested for covid-19, of whom 234 were positive. The rest tested negative and probably had other ailments, letting the researchers compare covid-19 with different diseases.

Covid-19 was distinctively disruptive. Although patients with symptoms who tested negative did increase hours of sleep and reduce the number of steps they took, these changes were modest and brief. In the covid-positive group, the effects were 2.5 times larger, and lasted for a month.

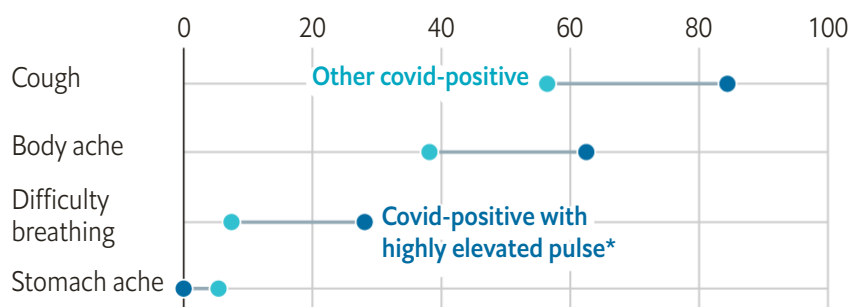
The most striking differences involved heart rates. An elevated pulse can leave tissue short of oxygen, causing lightheadedness or shortness of breath. Whereas the resting heart rates of people with non-covid ailments rose only slightly, those of covid-19 patients oscillated. In most cases—86% of people testing positive—pulses rose on average by 1.4 beats per minute (BPM), and then fell by 2.9BPM. As patients recovered, their hearts sped up again, to 0.5BPM above the starting point. They returned to normal after about 65 days.

However, the remaining 14% of people with covid-19—a share similar to the incidence of long covid found in some other studies—faced graver changes. Their pulses followed the same rise-fall-rise-fall pattern, over a similar time period. However, the impact was larger, with an average peak increase of 8.8BPM. Moreover, after 70 days their heart rates levelled out at 5.4BPM above the pre-covid pace. Their pulses stayed in that range until the study ended.

### Symptoms during early, acute phase of covid-19

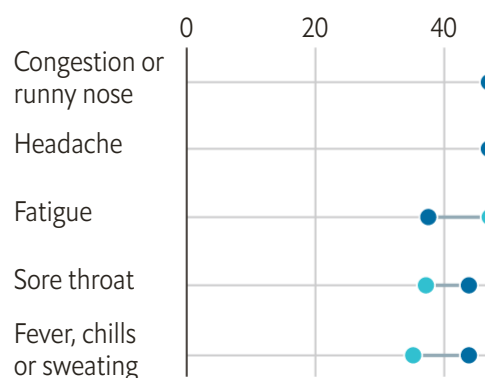
By change in heart rate 4-8 weeks into illness, % of group

#### Statistically significant differences



\*Average resting heart rate 4-8 weeks after onset of symptoms at least 5 beats per minute above pre-covid level

#### No statistically significant difference



Long-haulers often bemoan unending symptoms. Although the study did not track symptoms after the acute phase of covid-19, it confirms that the disease can affect vital signs for months. It also identifies risk factors for long covid: participants

with highly elevated pulses were unusually likely to suffer coughs, body aches and trouble breathing early in their illnesses.

Many young people think they do not need covid-19 vaccines because their risk is low. However, youth did not reduce participants' chances of having elevated pulses. Greater awareness of long covid might relieve such sceptics of this fallacy. ■

Source: "Assessment of prolonged physiological & behavioral changes associated with covid-19 infection", by J. M. Radin et al., JAMA Network Open, 2021

*This article appeared in the Graphic detail section of the print edition under the headline "Fitbit for purpose"*

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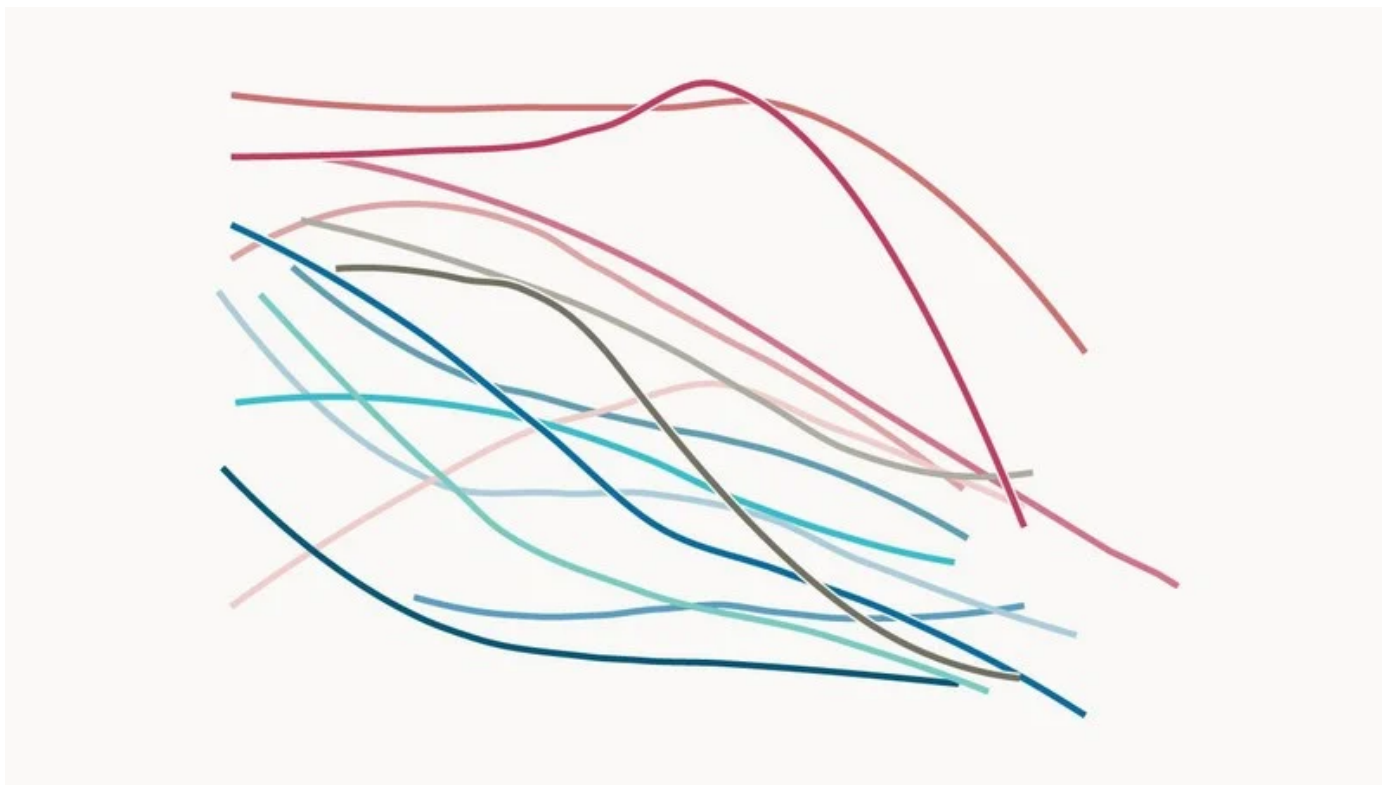
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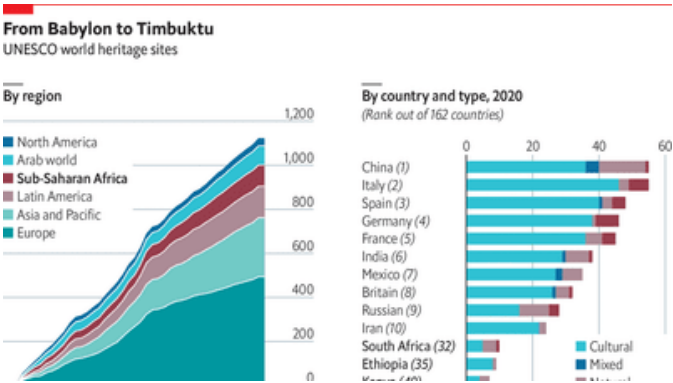
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